Alessandro Borghi

Control engineer & Applied mathematician

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Education

- **Ph.D. in Applied Mathematics**, TU Berlin, Germany *Oct 2021 – Present*
  Research focus: model order reduction, numerical linear algebra, optimization.
- **M.Sc. in Systems and Control**, TU Delft, Netherlands *2019 – 2021*
- **Double B.Sc. in Automation Engineering**,
  University of Bologna, Italy & Tongji University, China *2015 – 2019*

Professional Experience

- **Business Development Associate** - Sparksense, Zurich, Switzerland *Oct 2020 – Mar 2021*
  Developed B2B strategies for tech startups, improving market outreach and supporting product development.
- **R&D Intern** - Beckhoff Automation, Shanghai, China *Mar 2019 – Jul 2019*
  Designed and implemented active vibration compensation using input shaping in TwinCAT.
- **Intern** - CNH Industrial, Basildon, UK *Aug 2014*
  Assisted field engineers in vibration control testing for tractors.

Technical Skills

- **Programming:** MATLAB ▪ Python ▪ TwinCAT
- **Methods:** Control Systems ▪ Optimization ▪ Model Reduction ▪ Data-Driven Modeling ▪ Simulation
- **Languages:** Italian (native) ▪ English (C1, IELTS 7.5)

Selected Projects

- **Ph.D. Research – Model Order Reduction**
  Developing reduced-order modeling techniques for large-scale state-space dynamical systems.
- **M.Sc. Thesis – Koopman Subspace Identification**
  Data-driven modeling robust to measurement noise using subspace identification methods.
- **B.Sc. Thesis – Vibration Control of Flexible Transmission Systems**
  Simulation and PLC implementation of input shaping for vibration suppression.